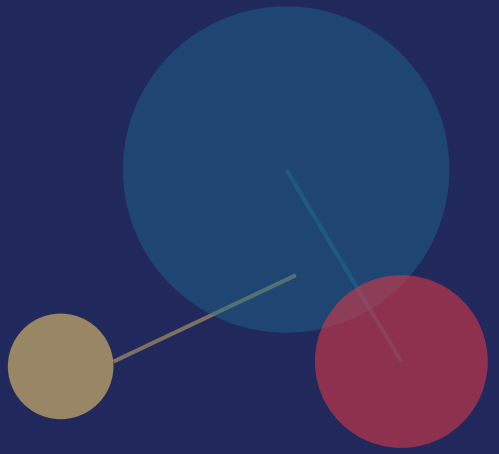


ADVERSARIAL GRAPH ML · ARCHITECTURE DEEP-DIVE

Diffusion-KATZE

How a discrete-time diffusion process injects adversarial nodes into a graph

Black-box · query-only · injection-only attack on GNN graph classifiers

The core idea: put edge topology — not just features — into the diffusion state

98.43%
PROTEINS GCN ASR

X + A
joint diffusion state

1 node
injection budget

SETUP

Threat Model — What the Attacker May Do

Black-box, query-only

Sees only output logits — never gradients or weights. Every gradient is a zeroth-order (CGE) estimate from queries.

Injection-only

The original graph is never mutated. The attacker appends new nodes with chosen features + edges; construction is a pure function.

Tiny budget

node_budget = 1 injected node; edges per node capped ($1 \le k \le$ avg-degree). A stealth constraint, not full connectivity.

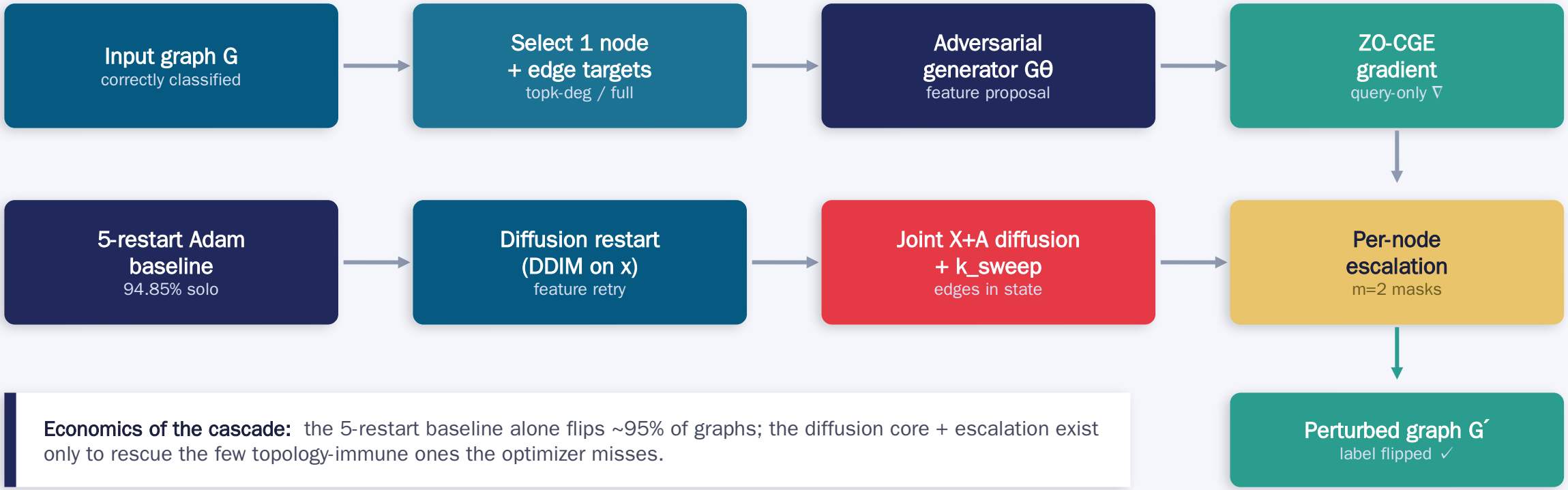
Goal: flip the label

Untargeted CW-margin objective. Success = the classifier changes its prediction on a previously-correct graph.

ARCHITECTURE

End-to-End Pipeline — A Cascade of Fallbacks

Each stage fires only when the previous one fails. Cheap methods first; the diffusion machinery is the last resort for the hard graphs.



Economics of the cascade: the 5-restart baseline alone flips ~95% of graphs; the diffusion core + escalation exist only to rescue the few topology-immune ones the optimizer misses.

Blue = data · Navy = generator / optimizer · Crimson = diffusion core · Gold = escalation · Green = success

COMPONENT A

Generator + Zeroth-Order Gradient

Adversarial generator Gθ

A small MLP maps noise → feature vector for the injected node. 5 random restarts (different lr / epochs) escape bad local optima.

CGE zeroth-order ∇

No weights, no autograd. Coordinate-wise Gradient Estimation perturbs the input ±σ and reads logit changes to estimate ∇.

feat_scale = 2√N

Injected features are amplified to ≈2√N. This single lever — not the optimizer — is what lifts ASR from ~30% to ~95%.

CW-margin loss

Carlini-Wagner margin (κ = −0.1) drives the true-class logit below the best other class — a smooth flip objective.

COMPONENT B

The Multi-Restart Baseline Ladder

Before any diffusion, 5 Adam restarts try to flip the graph. The best (lowest-margin) x_0 is cached as the diffusion anchor.

1. lr 5e-3 · 50 ep
2. lr 1e-2 · 50 ep
3. lr 2e-2 · 75 ep
4. lr 5e-3 · 100 ep
5. lr 1e-2 · 100 ep

Cache the best x_0 across restarts

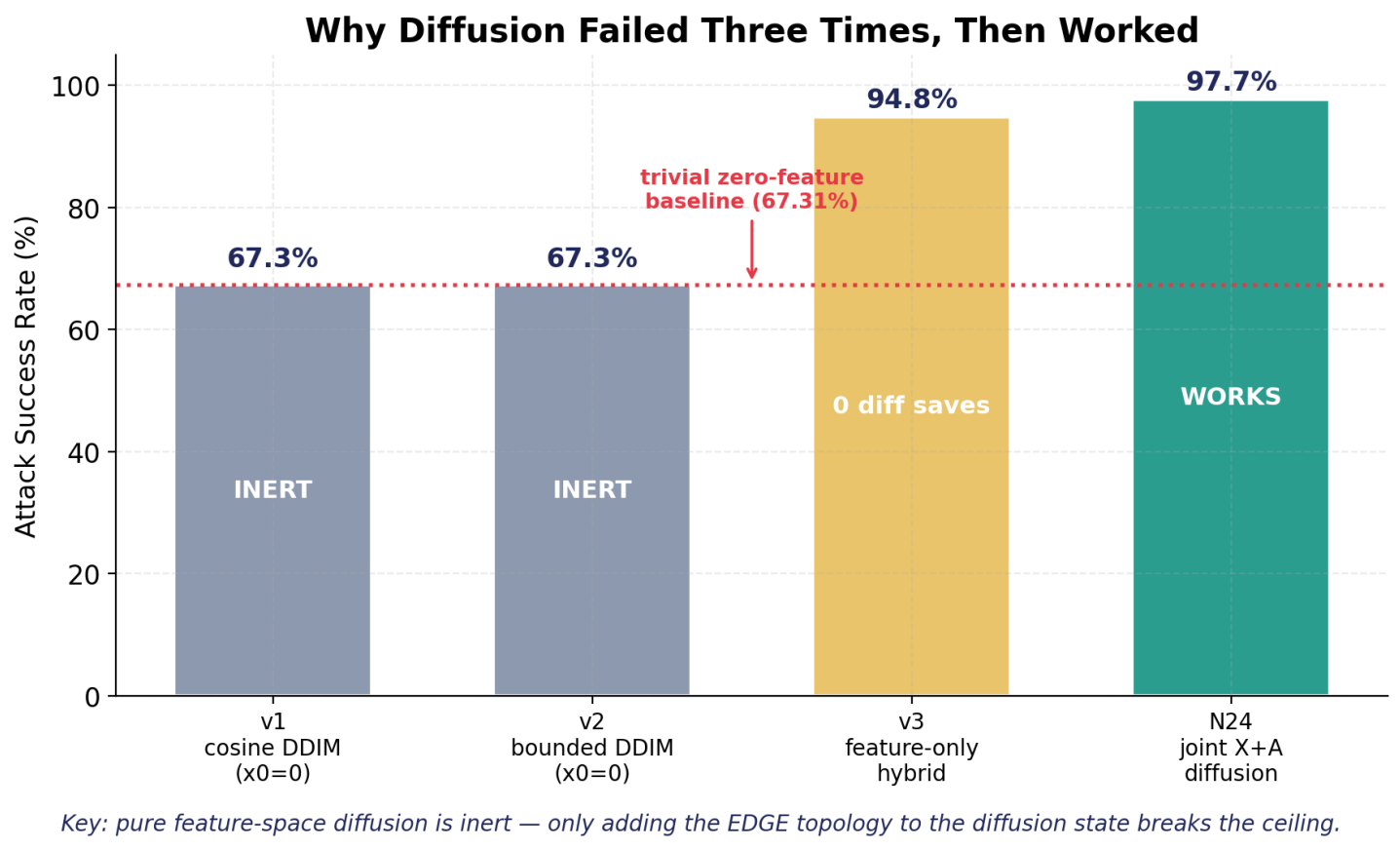
If any restart flips the graph → done (this alone solves ~95% of graphs). Otherwise the lowest-margin candidate becomes the anchor x_init that the diffusion process denoises toward.

Why an anchor matters

The early failure (N23) ran diffusion from $x_0 = 0$, where DDIM predicts $\hat{x}_0 = 0$ — testing only a zero-feature injection. Anchoring at the best baseline x_0 gives the score a meaningful place to denoise from.

WHY IT'S BUILT THIS WAY

N23 → N24: Why Feature-Only Diffusion Failed



Key: pure feature-space diffusion is inert — only adding the EDGE topology to the diffusion state breaks the ceiling.

THE ROOT CAUSE

v1–v3 diffused features only.
Under a fixed full-edge topology, GCN normalization + feat_scale saturate — feature content stops mattering. ~5% of graphs are topology-immune: no feature vector flips them.

Fix → make edges a variable.
Joint X+A diffusion adds a topological dimension the score can actually move.

67.3% → 97.7%
the moment edges entered the state

CORE IDEA

The Diffusion State is (X, A) — Not Just X

Feature-only diffusion

`state = x   (node features)`

- denoise x toward an adversarial feature vector
- edges fixed (full connect)
- **GCN norm absorbs it → inert**

67.31% — no gain over a zero-feature injection

Joint X + A diffusion

`state = (x, A_sv)   features + edges`

- denoise features AND search the edge mask
- edge cardinality bounded by budget k
- **adds a topological dimension → flips**

97.7%+ — the ceiling breaks

MECHANISM

The DDIM Engine + Adversarial-Margin Guidance

DISCRETE-TIME DDIM

Bounded linear schedule

$\bar{\alpha}$: 0.99 → 0.50 over T = 20 steps

Forward (noise the anchor)

$$x_T = \sqrt{\bar{\alpha}_T} \cdot x_{init} + \sqrt{1-\bar{\alpha}_T} \cdot \epsilon_0$$

Reverse step (predict-x_0 form)

$$\hat{x}_0 = (x_t - \sqrt{1-\bar{\alpha}_t}\cdot\epsilon) / \sqrt{\bar{\alpha}_t}$$

$$x_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\cdot\hat{x}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\cdot\hat{\varepsilon}$$

Bounded ($\bar{\alpha}$ never reaches 0 or 1): the anchor is only partially noised, so the trajectory stays near the adversarial manifold instead of wandering to pure noise.

ADVERSARIAL-MARGIN GUIDANCE

At each reverse step:

1. ZO-CGE estimates the score
   $s = \nabla_{\hat{x}_0}$ CW(model, graph)
2. steer the noise toward lower margin
   $\hat{\varepsilon} = \epsilon_0 - \lambda\cdot\sqrt{1-\bar{\alpha}_t}\cdot s$
   (λ = guidance = 2.0)
3. take the DDIM step → repeat

Over T = 20 steps the injection is denoised straight toward the decision boundary; the lowest-margin $\hat{x}_0$ is kept.

Queries replace gradients: the diffusion score is itself a zeroth-order estimate — the whole engine stays black-box.

MECHANISM

Searching the Edge Mask (the A in X+A)

Per-node edge mask

Each injected node i carries a binary mask edge_mask[i] ∈ {0,1}^N over original nodes, with $\|\text{mask}[i]\|_1 \leq k$ (cardinality budget).

Budget as upper bound

k is a ceiling, not a target. Greedy search keeps only positive-value edges, so the attack uses the fewest edges that still flip.

k_sweep on stuck graphs

If the default k fails, retry with k ∈ {0.25, 0.5, 1.5, 2}× — different cardinalities suit different graph sizes.

2-swap refinement

After top-k projection, try top-K-in × top-K-out swaps (K^2=16) to escape coupled-edge optima. Same ASR, 35% faster.

Per-node escalation (m=2)

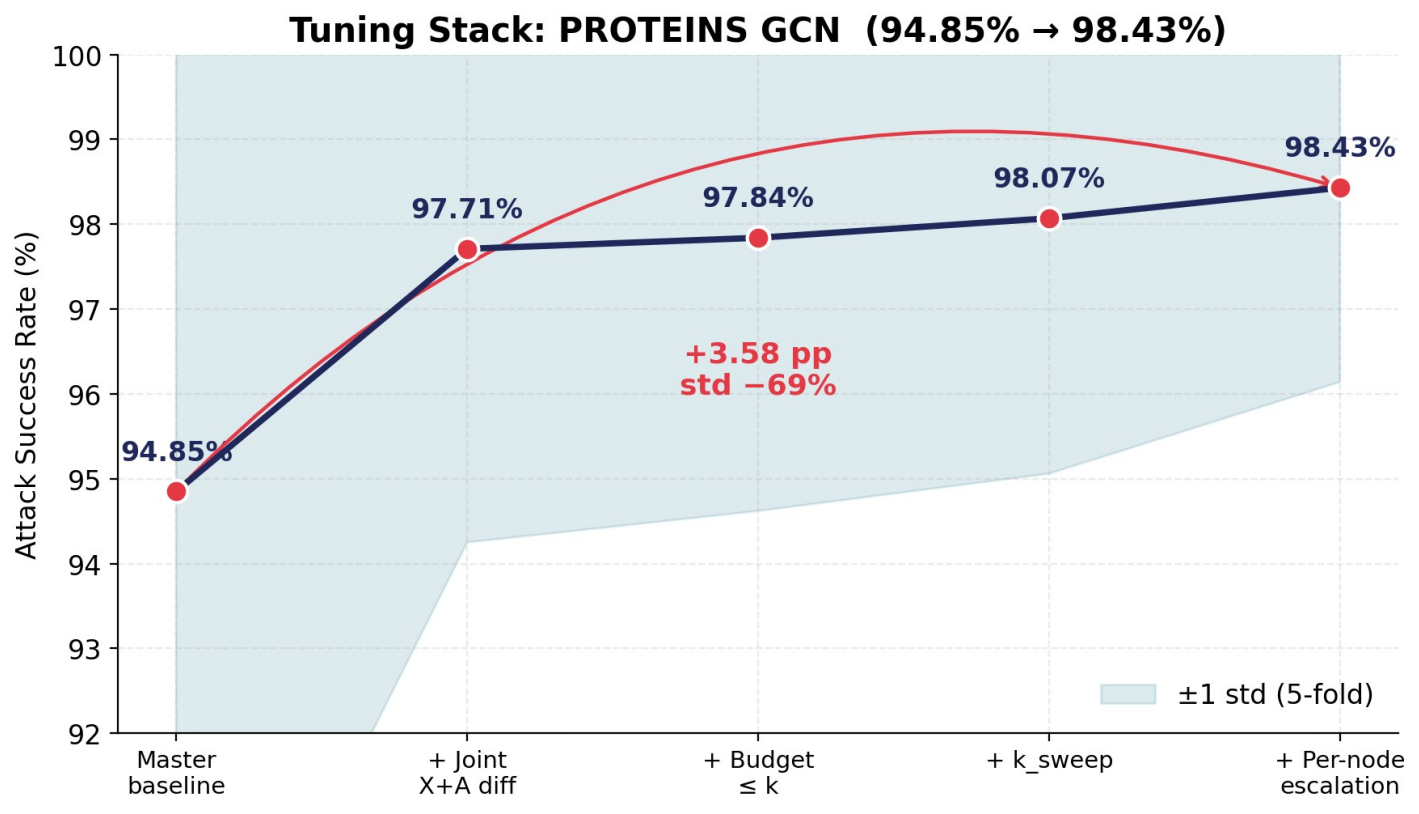
Add a 2nd injected node with an INDEPENDENT mask. Shared masks make duplicate hubs that interfere (−1.85 pp); independent masks add +0.36 pp.

Multi-seed

2 random seeds per diffusion attempt — cheap variance reduction on the stochastic edge search.

PUTTING IT TOGETHER

When Each Tier Fires (94.85% → 98.43%)



1. Baseline

5-restart Adam + ZO-CGE

2. Joint X+A diffusion

bounded DDIM + edge search

3. Budget ≤ k

edges as upper bound

4. k_sweep

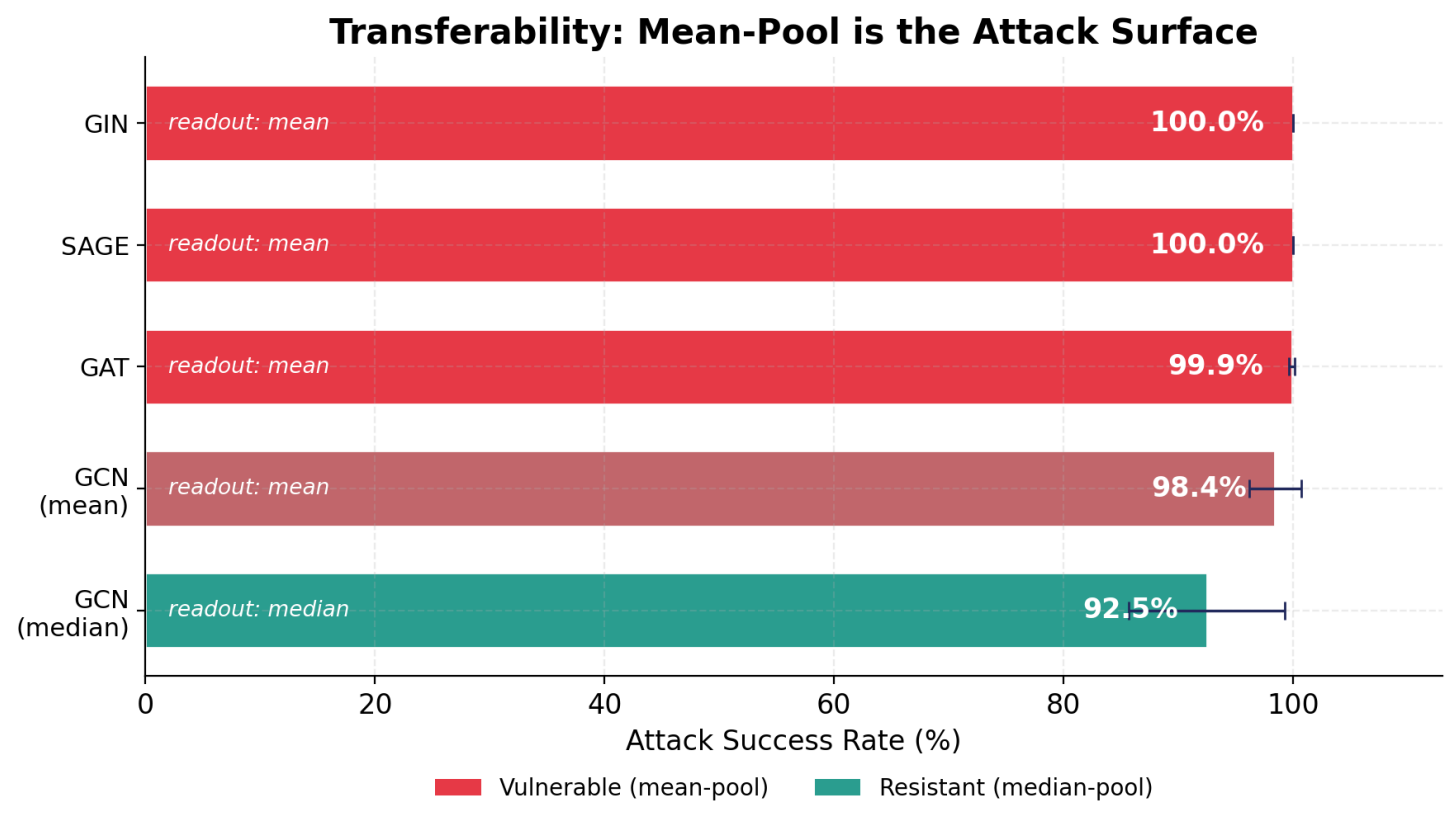
retry stuck graphs at other k

5. Per-node escalation

m=2, independent masks

DESIGN INSIGHT

The Readout — Not the Convolution — is the Attack Surface



Mean-pool falls GIN (sum), GAT (attention), SAGE (sampling) all use mean-pool readout — and all hit 99.9–100% ASR. Changing the conv is no defense.

Median-pool resists Only a median / trimmed-mean readout ignores the injected outlier node — dropping ASR to 92.46%. Defend the pooling.

Diffusion earns its keep here On undefended models a baseline suffices; against median-pool, diffusion is the difference between ~63% and 92%.

RESULTS

Fair, Budget-Matched Comparison (PROTEINS GCN)

All methods at node_budget = 1, 5-fold. The original 54.26% → 98.43% was NOT budget-matched; under one identical budget the gap is honest.

Method	GZOO budget (1 edge)	N24 budget (full)
GZOO-origin (fs=1.0)	29.65 ± 12.04	75.25 ± 4.46
gzoo-katze (fs=auto_x2)	92.31 ± 5.42	94.85 ± 7.44
diffusion-katze (N24)	**95.38 ± 3.87**	**98.43 ± 2.29**
GZOO-paper (cited)	54.26	n/a

+65.7 pp

diffusion-katze vs GZOO-origin
at the SAME 1-edge budget

feat_scale

the dominant lever — fs=1.0→auto_x2
moves ASR far more than the optimizer

edges matter

every method gains from full edges;
topology is a real budget axis

Architecture in Three Sentences

1. **Cascade of fallbacks** — Cheap 5-restart Adam + ZO-CGE solves most graphs; the diffusion engine is the last resort for the hard, topology-immune ones.

2. **Topology in the state** — The decisive design choice is making edges part of the diffusion state (X+A). Feature-only diffusion is provably inert under mean-pool GCNs.

3. **Black-box throughout** — Even the diffusion score is a zeroth-order estimate — no gradients, no weights, 1 injected node. Defend the readout, not the convolution.

Diffusion-KATZE · autoresearch-katze · PROTEINS / GNN victims